MCU Connections

U2
STM32C542CCT6

RST_BNO	1	PE2	VDD	48	3.3V
INT_BNO	2	PC13 Low current/freq	VSS	47	
	3	PC14-OSC32_IN Low current/freq	PB9	46	DEBUG
	4	PC15-OSC32_OUT Low current/freq	PB8	45	CAN_RX
OSC_IN	5	PH0-OSC_IN	PH2-BOOT0	44	
OSC_OUT	6	PH1-OSC_OUT	PB7	43	CAN_TX
nRST	7	NRST	PB6	42	CS_BNO
	8	VREF- Not using DAC/ADC	PB5	41	CS_BMI
	9	VREF+ GND VREF+?	PB4	40	INT_BMI
PWM5	10	PA0	PB3	39	SOL3A
PWM6	11	PA1	PA15	38	SOL3B
PWM7	12	PA2	PA14	37	SWCLK
PWM8	13	PA3	VDD	36	3.3V
MOSI	14	PA4	VSS	35	
SCLK	15	PA5	PA13	34	SWDIO
MISO	16	PA6	PA12	33	SDA
SOL1A	17	PA7	PA11	32	PWM1
SOL1B	18	PB0	PA10	31	PWM2
SOL2A	19	PB1	PA9	30	PWM3
SOL2B	20	PB2	PA8	29	PWM4
	21	PB10	PB15	28	SOL4A
	22	VCAP	PB14	27	SOL4B
	23	VSS	PB13	26	SOL5A
	24	VDD	PB12	25	SOL5B

* Assuming no ADC

C11 0.1u (3.3V)
R5 10k (PH2-BOOT0)
C2 0.1u (nRST)
R6 10k (3.3V, SDA)
C10 0.1u (3.3V)
3.3V — R1 10k — SCL
C1 2.2u
C5 0.1u
C6 4.7u

Categorized Designs

- MOSFET_driver — File: untitled.kicad_sch
- Sensors_Telemetry — File: Sensors_Telemetry.kicad_sch
- Connector_Signals — File: connector_signals.kicad_sch
- Voltage Conv-Protec-Monitoring — File: Voltage_conv_prot.kicad_sch

Crystal Oscillator

Y1
crystal 3225 gnd 24
ECS-240-8-37B-CKY-TR
R2 0
OSC_IN — C7 8p — C8 8p — OSC_OUT
NEW CAP VALUES?

1206 package size

5.2V_pre_sense — C12 22u
12V_pre_sense — C13 22u — C14 22u

Test Points, LEDs

Footprint fix, it's backwand

TP1 GND_TP
TP2 5.2V_TP
TP3 12V_TP
TP4 3.3VTP
D1 ORANGE — R7 120 (DEBUG)
D2 BLUE — R8 1k (12V)
D3 GREEN — R9 320 (5.2V)
D4 RED — R10 120 (3.3V)
12V 5.2V 3.3V

10mA all

CAN Transceiver

U1
TCAN1044AEVDRQ1

5.2V	3	VCC	CANH	7	CAN_H
3.3V	5	VIO			
CAN_TX	1	TXD			
	8	STB	CANL	6	CAN_L
CAN_RX	4	RXD	GND	2	

C3 0.1u
C4 0.1u
R3 60
R4 60
C9 0.1u

CR1
ESD2CAN24DBZRQ1

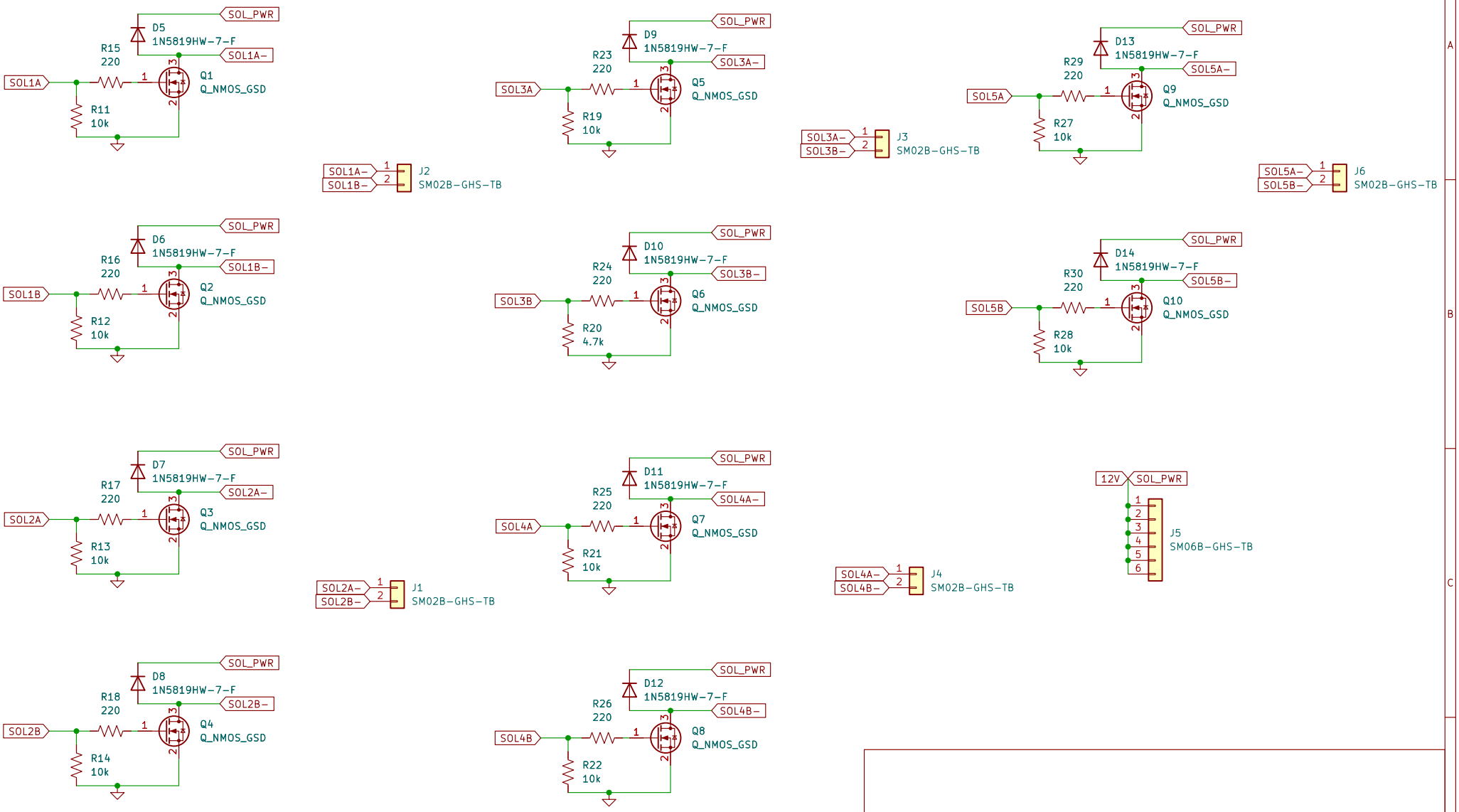
Place TVS close to connector

Purdue ROV
Sheet: /
File: X19-Control-Board.kicad_sch

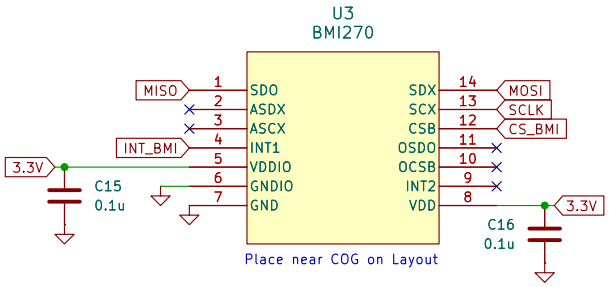
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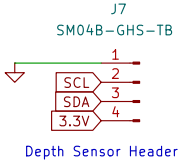
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KiCad E.D.A. 10.0.5-unknown		Id: 1/5

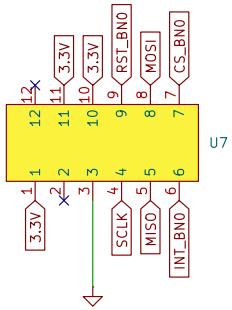
3 is the backup

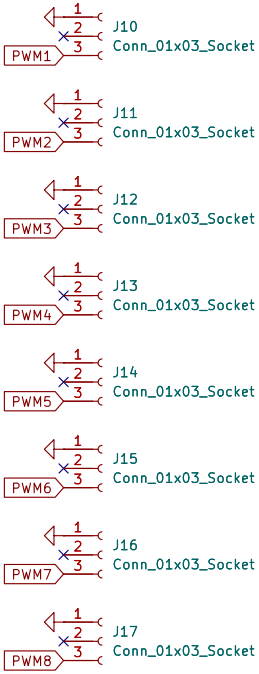


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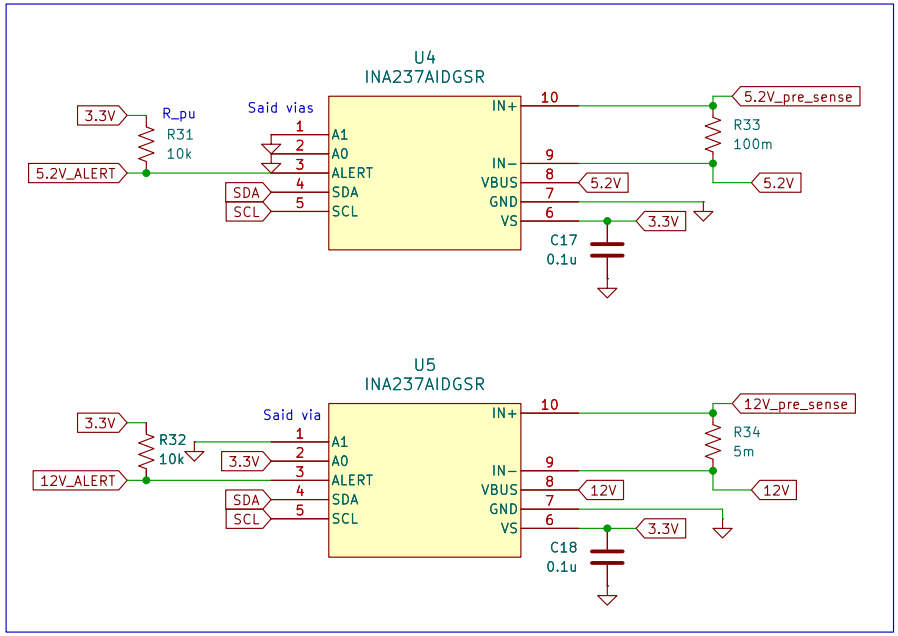


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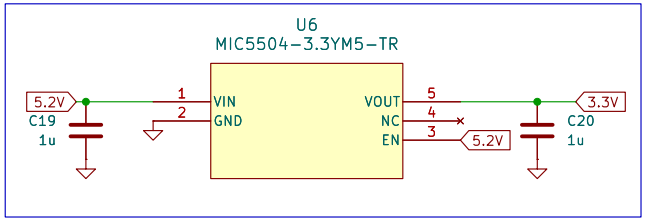
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KiCad E.D.A. 10.0.5-unknown		Id: 5/5

Power Monitoring 5.2V, 12V rails



LDO – Voltage Conversion



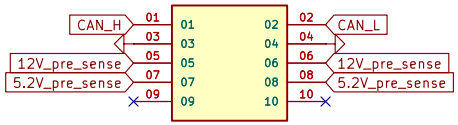
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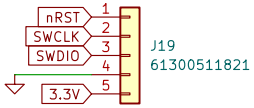
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THIS PINOUT IS NOT FINAL

J18
IPS1-105-01-L-D

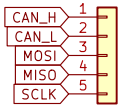


Break Out SWD



J19
61300511821

Break Out Communication



J20
61300511821

Place close to MCU

Connector added for debug if there is space

Sheet: /Connector_Signals/ File: connector_signals.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.5-unknown		Id: 7/5